RSLogix 5000
Getting Started Switch1 → LED1

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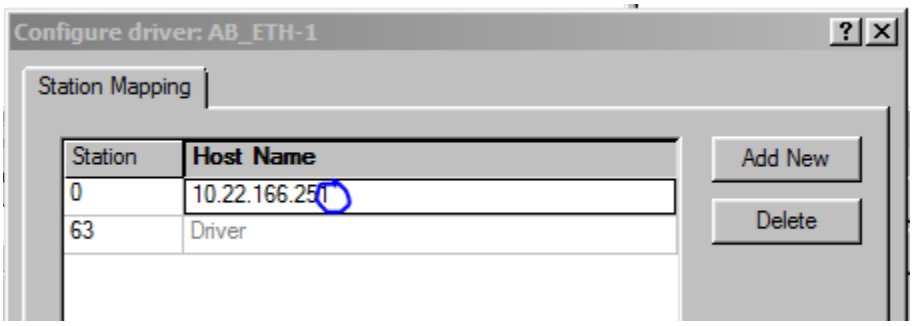
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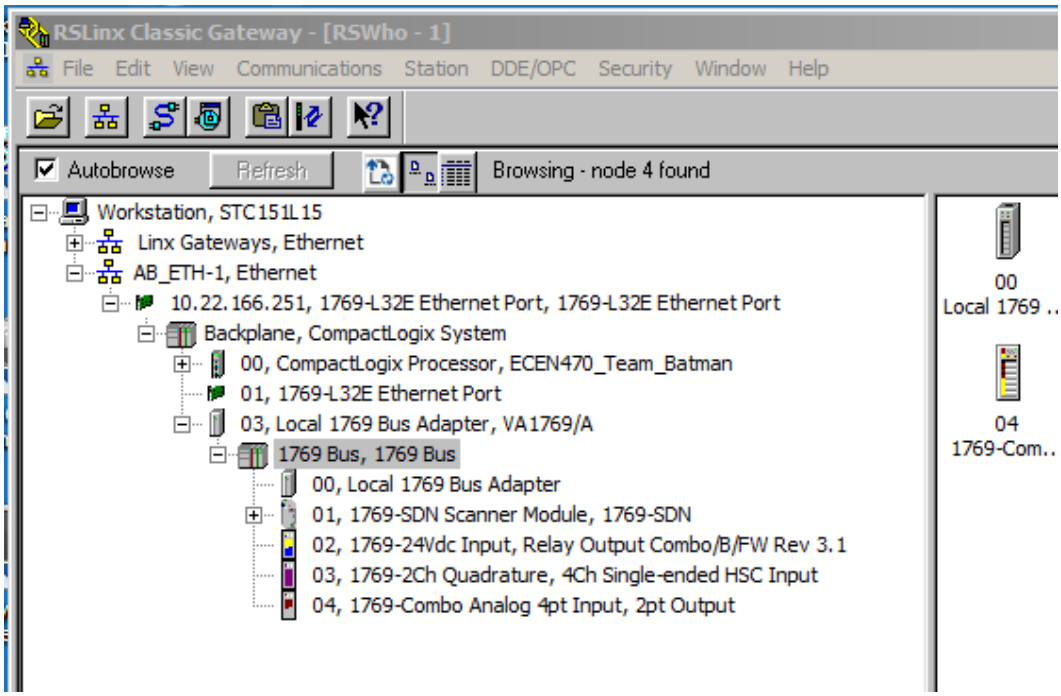
Setup Communication link with RSLinx Classic

1. Start RSLinx Classic, select Start - All Programs - Rockwell Software - RSLinx Classic
2. Select Communications - Configure Drivers…
3. Select, Stop, and Delete any existing drivers listed in Configured Drivers
4. Under Available Driver Types, select Ethernet devices - Add New… - OK
5. Type the IP address written on the PLC in the Host Name field:



6. Select OK - Close to close the two dialog boxes

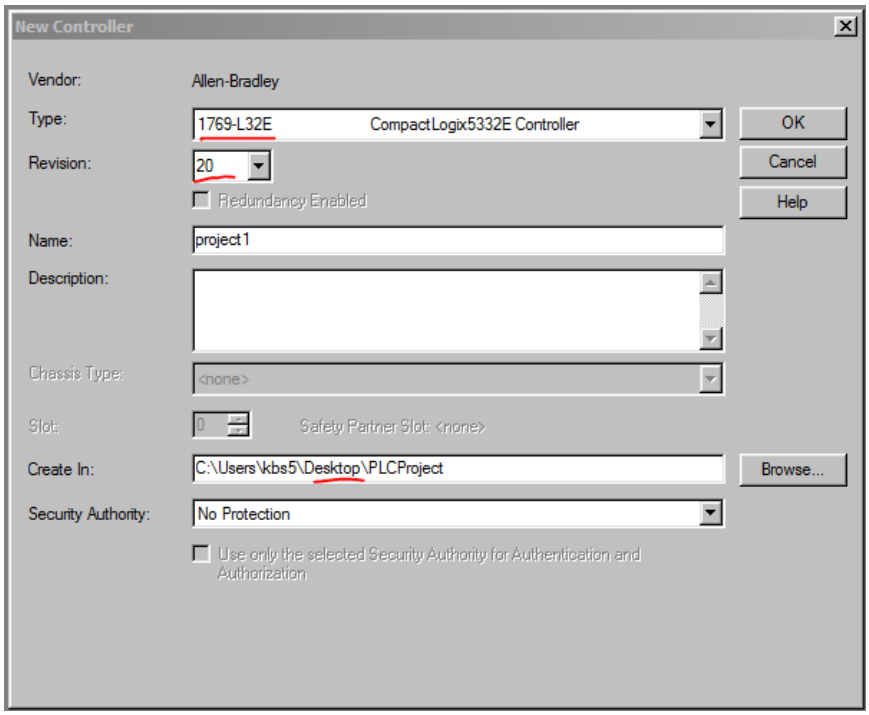
   The PLC should now be visible:



7. Select File - Exit to close RSLinx Classic Gateway

To Create a New RSLogix 5000 Project

1. To start RS Logix 5000
    a. Select Start - All Programs - Rockwell Software -RSLogix 5000
2. Select File - New…
3. Enter the following
    a. Type: 1769-L32E
    b. Revision: 20
    c. Name: project_1
    d. Create In: ..\Desktop\PLCProject

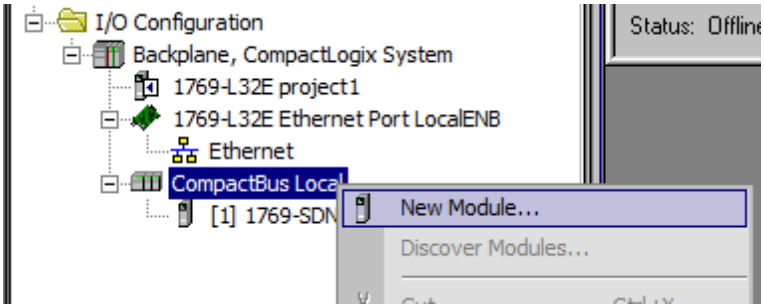


4. Select **OK**
5. Click **Yes** to create the directory

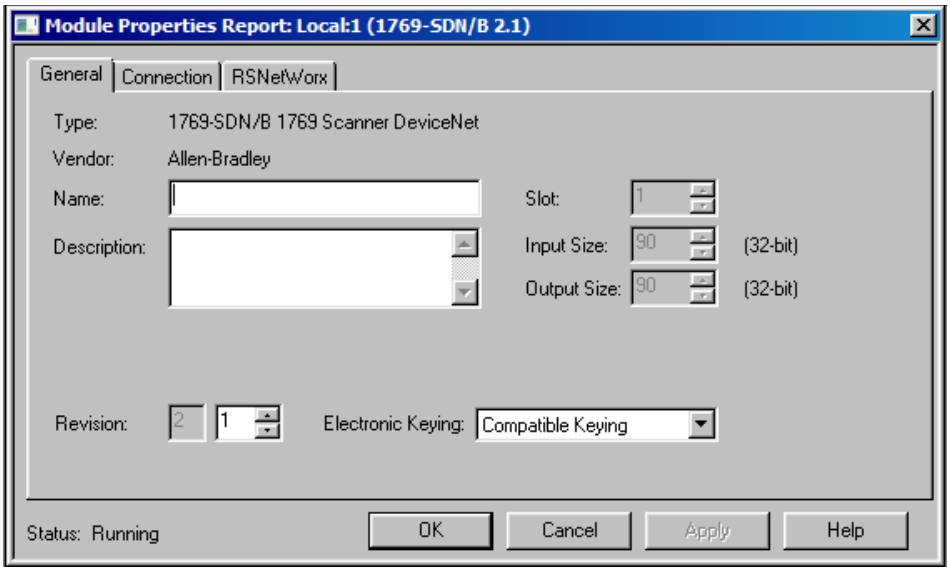
To Add I/O Modules

1. Locate the Controller Organizer section
2. **Expand** I/O Configuration
3. **Expand** Backplane, CompactLogix System

   Add the 1769 Scanner DeviceNet module
4. **Right**-Click CompactBus Local and select **New Module**



5. Select **1769-SDN** "1769 Scanner DeviceNet"
6. Click **Create**
7. Select **2** for the Major Revision
8. Click **OK**

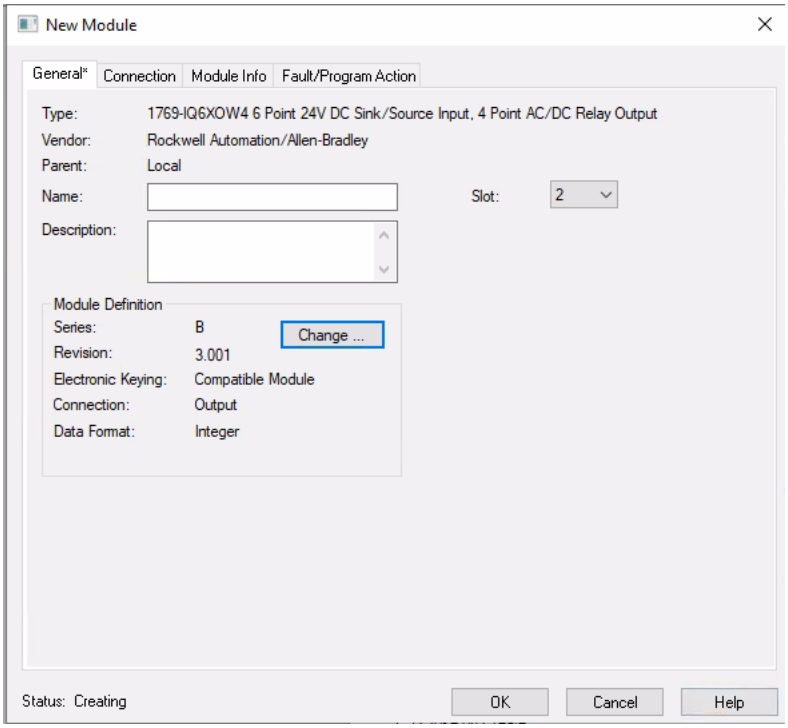


9. Select OK to close the New Module

   We will now repeat the process to add the second module.

   <u>Add the Digital I/O module</u>

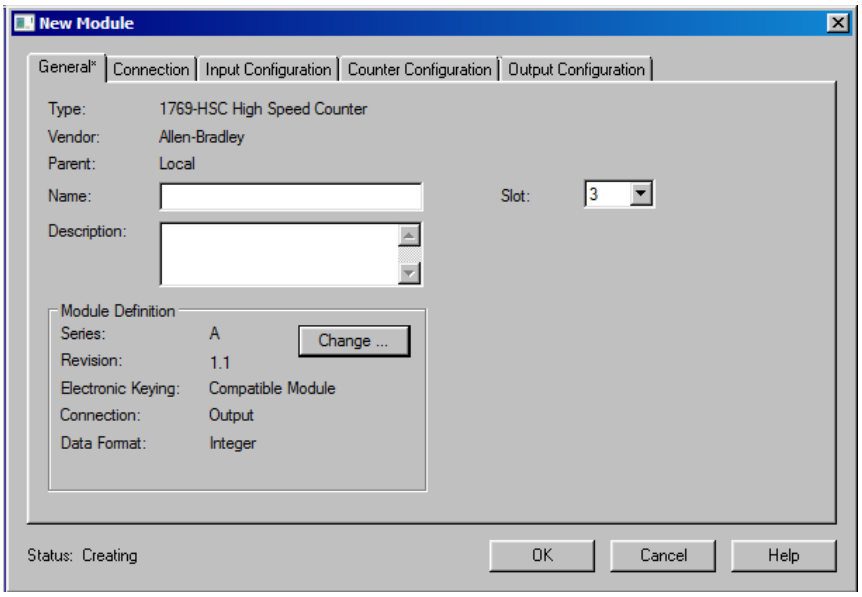
10. **Right**-Click CompactBus Local and select **New Module**…
11. Select **1769-IQ6XOW4** "6 Point 24V DC Sink/Source Input. 4 Point AC/DC Rel… "
12. Click **Create**
13. Under Module Definition, select **Change.**
14. Change the Revision: to **3** and 001.
15. Click **OK**.
16. Click **Yes** to confirm changing the module definition.



17. Click **OK** to close the New Module

    Add the High Speed Counter

18. Select Specialty Modules in the Module Type Category Filter.
19. Select **1769-HSC** "High Speed Counter"
20. Click **Create**.
21. Under Module Definition, select **Change.**
22. Change Series to **A**.
23. Set the Revision to **1** and 001.
24. Click **OK**.
25. Click **Yes** to confirm changing the module definition.

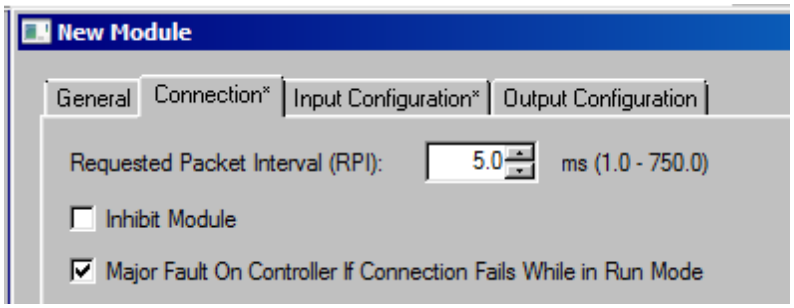


26. Click **OK**.

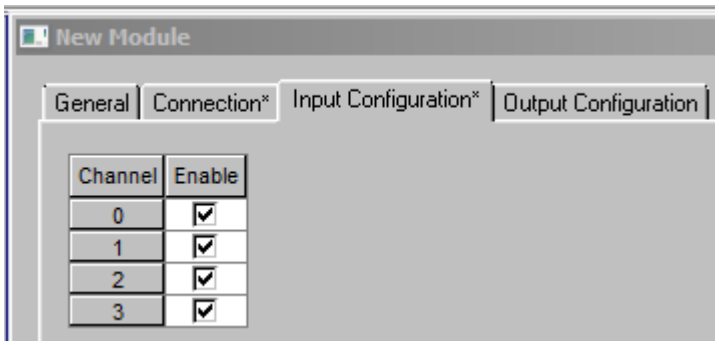
Add the Analog I/O Module

27. Under Analog Modules, select **1769-IF4XOF2** "4 Channel Input/2 Channel Output Low Resolution…"
28. Click **Create**.
29. Under Module Definition, select **Change.**
30. Change the Revision: to **1** and 001.
31. Click **OK**.
32. Select Connection Tab
33. Set (PRI) = 1.0 ms.

Note, 1.0 ms is only needed when controlling fast systems.

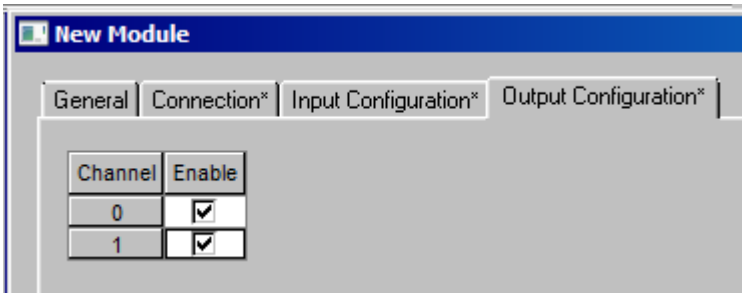


34. Select Input Configuration tab
35. Enable all the channels.



36. Select Output Configuration tab.

37. Enable all the channels.

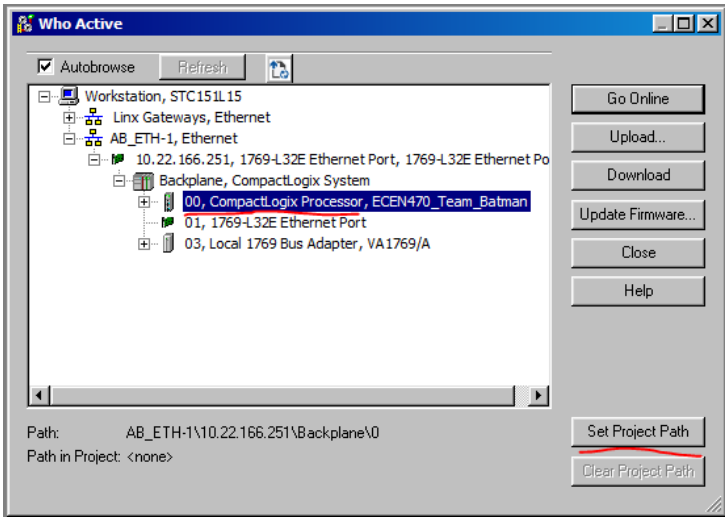


38. Select **OK** to close.

39. Select Close to close the Select Module Type

Set Project Path to the PLC

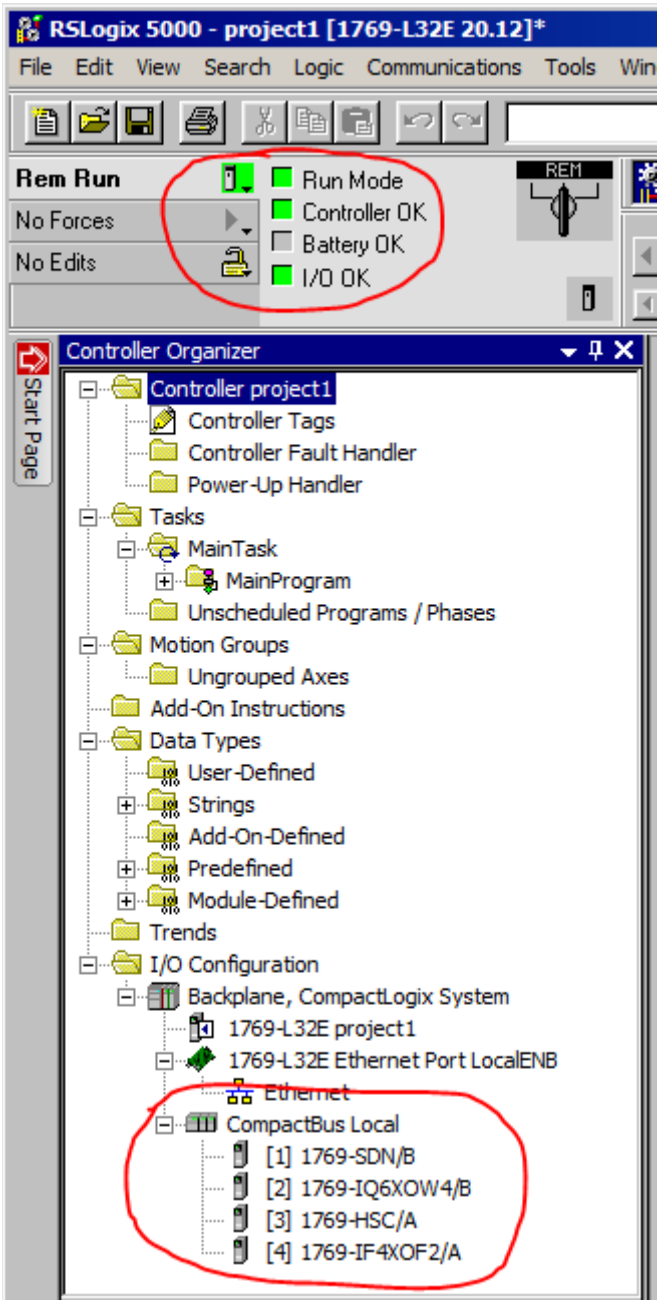
1. Click the Communication Tab
2. Select Who Active
3. Expand the AB_ETH-1, Ethernet until you reach **00, CompactLogix Processor**.
4. Select **00, CompactLogix Processor**.
5. Click **Set Project Path**.



Download new project to the PLC

1. With the Who Active window still open, select **Download**.
2. Select **Download** in the Download window.
3. Select **Yes** to Change controller mode back to Remote Run, if prompted.
    a. Otherwise, select **Rem Prog – Run Mode**.

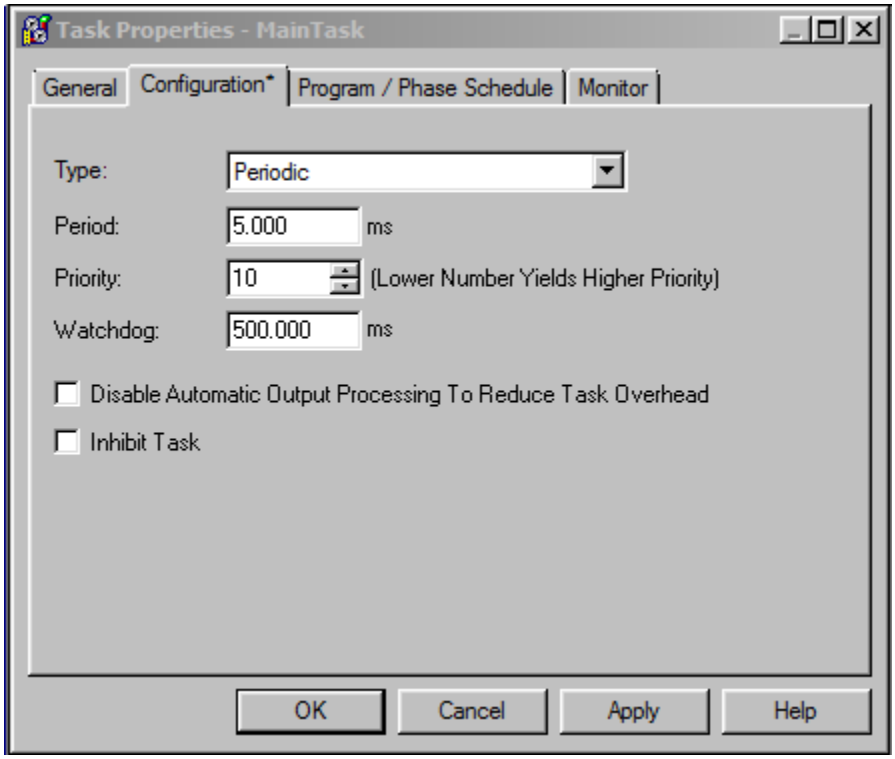
The Run Mode indicator should now be green and the CompactBus Local should be as shown



Set Program Modes

1. Select Rem Run - **Go Offline**.
2. Expand the Tasks folder in the Controller Organizer.
3. **Double**-Click MainTask folder.
4. Select Configuration Tab
5. Select Type to **Periodic** and the Period to **5.000** ms.

   Note: 5.00 ms is only when controlling fast systems.



6. Select **OK** to close.
7. Expand the MainTask folder and the MainProgram folder.
8. Right-click MainProgram and select New Routine.
9. In the Name field, enter myFirstRoutine.
10. In the Type filed, select Function Block Diagram.
11. Click **OK**.
12. Double-click the MainProgram folder.
13. Select the Configuration tab.
14. In the Main field, select myFirstRoutine.
15. Click **OK**.

16. Right-click MainRoutine and select Delete.
17. Click **Yes** to confirm the deletion.

View I/O Module Data

1. Select Communications – Download.
2. Select Download.
3. Select Yes to change controller mode back to Remote Run.
4. Expand **Controller project_1** in the Controller Organizer.
5. **Double-**click Controller Tags.
6. Select the Monitor Tags tab at the bottom.
7. Expand **Local:2:I.**
8. Expand **Local:2:I.Data.0**.
   Note the value of **Local:2:I.Data.0**.
9. Toggle Switch one on the PLC Trainer.
   Verify that the **Local:2:I.Data.0** tag toggles between 0 and 1 with the switch position.
10. Expand **Local:2:O**
11. Expand **Local:2:0.Data**.
12. Change the value to **1** for tag Local:2:O.Data.0.
    The LED 1 should turn on in the PLC Trainer.

Create Local Tags and Alias them to Controller Tags

1. Expand **Tasks – MainTasks – MainProgram**.
2. **Double-**click **Program Tags**.
3. Select **Edit Tags** tag at the bottom.
4. Enter the name **Switch1** in the name column for the new tag.
   a. Click in the **Alias For** field and click the down arrow that appears.
   b. Expand **Local:2:I**
   c. Find Local:2:I.Data and click on the down arrow.
   d. Select bit 0.
      Local:2:I.Data.0 should be in the Alias For cell and Switch1 is now aliased to this address.
5. Repeat this process to create a tag named **LED1** and alias it to **Local:2:O.Data**.0.

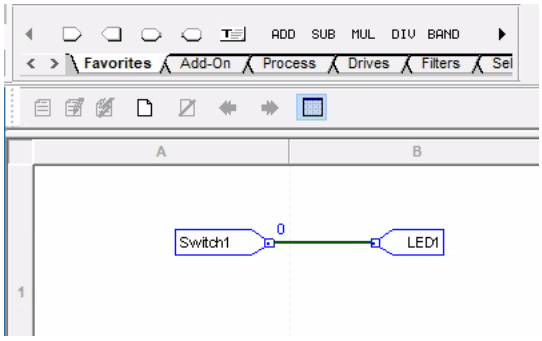
View and Change Local Tag Values

1. Select **Monitor Tags** tab.
2. Toggle Switch one on the PLC Trainer.
   Verify that the value of Switch1 changes between 0 and 1.
3. Click the value filed for LED1 and change the value between 0 and 1.
   Verify that LED1 is off when the value is 0 and on when the value is 1.

Program Switch1 to Control the State LED1

1. Select **File – Save**.
2. Click **Yes** to upload tag values before saving the project?
3. Select **Communications – Go Offline**.

4. Click **Yes** to confirm.
5. **Double**-click myFirstRoutine.
6. Right-click the white space and select Add Element.
7. Select IREF (Input Reference) and select OK.
8. Double-click the IREF symbol.
9. Select the down arrow to browse for the Switch1 tag.
10. Double-click Switch1 and hit enter.
11. Right-click the white space and select Add Element.
12. Select OREF (Output Reference) and select OK.
13. Double-click the OREF symbol.
14. Select the down arrow to browse to the LED1 tag.
15. Double-click LED1 and hit enter.
16. Click and drag from the IREF tab to the OREF tab to connect them.



17. **Click** Verify Routine icon  .
    The Error window will open, verify that there are 0 errors and 0 warnings.
18. Select Communications – Download.
19. Select Download.
20. Select Communications – Run Mode.
21. Click Yes to confirm.
    Verify that LED1 turns on and off as Swith1 is toggled on and off.
22. Select File – Save.
23. Select File – Exit.

End of Lab